

DSC540Week1&2

March 15, 2025

1 Activity 1.01

Create a Jupyter notebook where you create a list, iterate over the list and sort your results, generate random numbers, add to the list, and then print your results.

```
[14]: import random

# empty list
my_list = []

# Add random numbers to the list
for _ in range(20):
    my_list.append(random.randint(1, 100))

# Print the unsorted list
print("Unsorted list:", my_list)

# Sort the list
my_list.sort()

# Print the sorted list
print("Sorted list:", my_list)
```

Unsorted list: [72, 78, 57, 49, 33, 48, 30, 59, 5, 13, 44, 86, 78, 68, 64, 86, 55, 42, 50, 96]

Sorted list: [5, 13, 30, 33, 42, 44, 48, 49, 50, 55, 57, 59, 64, 68, 72, 78, 78, 86, 86, 96]

```
[5]: pip install openpyxl
```

Requirement already satisfied: openpyxl in /opt/anaconda3/lib/python3.12/site-packages (3.1.2)

Requirement already satisfied: et-xmlfile in /opt/anaconda3/lib/python3.12/site-packages (from openpyxl) (1.1.0)

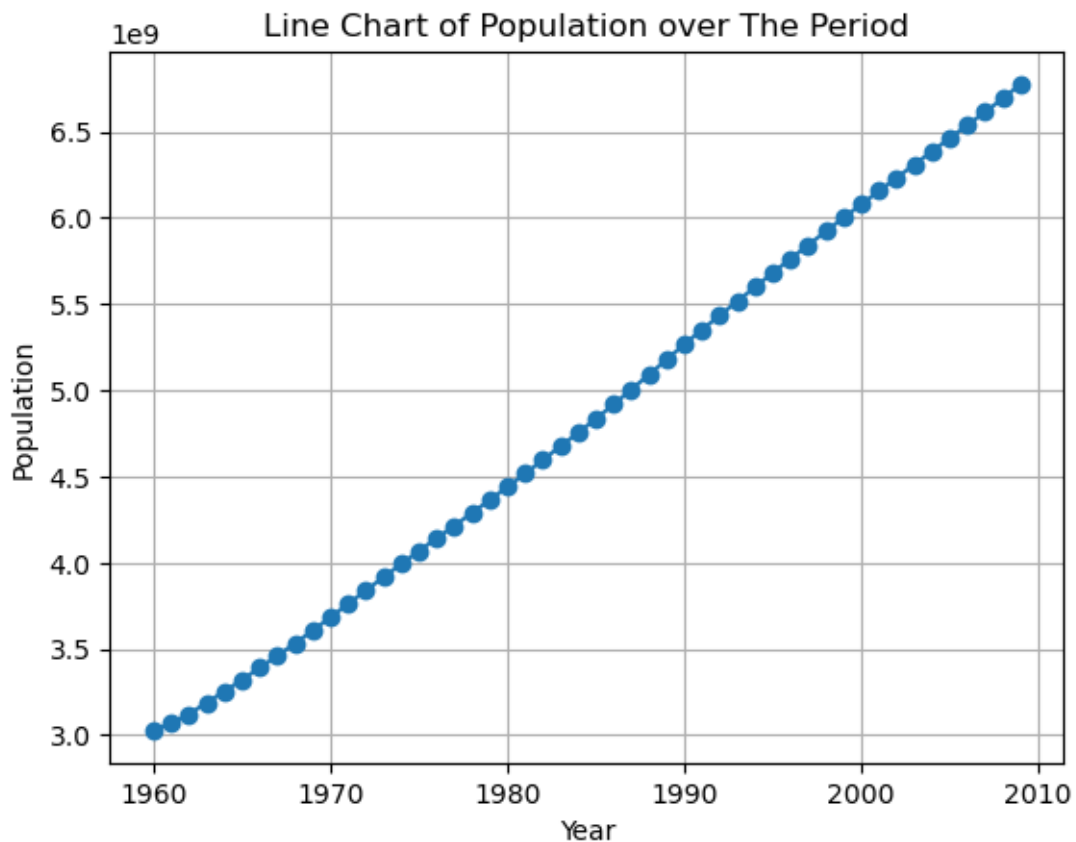
Note: you may need to restart the kernel to use updated packages.

Loading the required libraries and reading the data from world population file to plot it over the period.

```
[25]: import openpyxl
import matplotlib.pyplot as plt
import pandas as pd
import io

# Load the world population file
df_world_pop = pd.read_excel('world-population.xlsm',
    ↳sheet_name="world-population")

plt.plot(df_world_pop['Year'], df_world_pop['Population'], marker='o',
    ↳linestyle='-')
plt.xlabel('Year')
plt.ylabel('Population')
plt.title('Line Chart of Population over The Period')
plt.grid(True)
plt.show()
```



```
[16]: # Get numbers from my_list generated at first step and provide numbers those
    ↳are divisble by 3
```

```
list_with_divisible_by_3 = [a for a in my_list if a % 3 == 0]
list_with_divisible_by_3
```

```
[16]: [30, 33, 42, 48, 57, 72, 78, 78, 96]
```

```
[18]: # get the length difference between my_list and list_with_divisible_by_3
```

```
len_my_list = len(my_list)
len_divisible_list_by_3 = len(list_with_divisible_by_3)
difference = len_my_list - len_divisible_list_by_3
difference
```

```
[18]: 11
```

```
[27]: NUMBER_OF_EXPERIMENTS = 20
      difference_list = []
      for i in range(0, NUMBER_OF_EXPERIMENTS):

          list_with_divisible_by_3 = [a for a in my_list if a % 3 == 0]

          len_my_list = len(my_list)
          len_divisible_list_by_3 = len(list_with_divisible_by_3)
          difference = len_my_list - len_divisible_list_by_3
          difference_list.append(difference)
      difference_list
```

[illegible]

```
[29]: # Get the average for length difference

avg_diff = sum(difference_list) / float(len(difference_list))
avg_diff
```

```
[29]: 11.0
```

2 Activity 1.02

2.0.1 Below is multiline text provided in exercise though we can use our own text as well but do not want to create lot of confusion since using the provided text.

```
[37]: multiline_text= """It is a truth universally acknowledged, that a single man in
    ↪possession of a good fortune, must be in want of a wife.
    However little known the feelings or views of such a man may be on his first
    ↪entering a neighbourhood, this truth is so well fixed in the minds of the
    ↪surrounding families, that he is considered the rightful property of some
    ↪one or other of their daughters.
    "My dear Mr. Bennet," said his lady to him one day, "have you heard that
    ↪Netherfield Park is let at last?"
    Mr. Bennet replied that he had not.
    "But it is," returned she; "for Mrs. Long has just been here, and she told me
    ↪all about it."
    Mr. Bennet made no answer.
    "Do you not want to know who has taken it?" cried his wife impatiently.
    "You want to tell me, and I have no objection to hearing it."
    This was invitation enough.
    "Why, my dear, you must know, Mrs. Long says that Netherfield is taken by a
    ↪young man of large fortune from the north of England; that he came down on
    ↪Monday in a chaise and four to see the place, and was so much delighted with
    ↪it, that he agreed with Mr. Morris immediately; that he is to take
    ↪possession before Michaelmas, and some of his servants are to be in the
    ↪house by the end of next week."
    "What is his name?"""
```

Get the type of multiline data

```
[40]: type(multiline_text)
```

```
[40]: str
```

Get the length of the multiline text

```
[43]: len(multiline_text)
```

```
[43]: 1218
```

Cleaning the multiline text statement

```
[46]: multiline_text = multiline_text.replace('\n', "")

multiline_text
```

```
[46]: 'It is a truth universally acknowledged, that a single man in possession of a
good fortune, must be in want of a wife.However little known the feelings or
views of such a man may be on his first entering a neighbourhood, this truth is
so well fixed in the minds of the surrounding families, that he is considered
the rightful property of some one or other of their daughters."My dear Mr.
Bennet," said his lady to him one day, "have you heard that Netherfield Park is
let at last?"Mr. Bennet replied that he had not."But it is," returned she; "for
Mrs. Long has just been here, and she told me all about it."Mr. Bennet made no
answer."Do you not want to know who has taken it?" cried his wife
impatiently."You want to tell me, and I have no objection to hearing it."This
was invitation enough."Why, my dear, you must know, Mrs. Long says that
Netherfield is taken by a young man of large fortune from the north of England;
that he came down on Monday in a chaise and four to see the place, and was so
much delighted with it, that he agreed with Mr. Morris immediately; that he is
to take possession before Michaelmas, and some of his servants are to be in the
house by the end of next week.""What is his name?'
```

Removing special chars and punctuation and other unwanted characters from text

```
[52]: cleaned_multiline_text = ""
for char in multiline_text:
    if char == " ":
        cleaned_multiline_text += char
    elif char.isalnum(): # using the isalnum() method of strings.
        cleaned_multiline_text += char
    else:
        cleaned_multiline_text += " "

cleaned_multiline_text
```

```
[52]: 'It is a truth universally acknowledged  that a single man in possession of a
good fortune  must be in want of a wife However little known the feelings or
views of such a man may be on his first entering a neighbourhood  this truth is
so well fixed in the minds of the surrounding families  that he is considered
the rightful property of some one or other of their daughters  My dear Mr
Bennet  said his lady to him one day  have you heard that Netherfield Park is
let at last  Mr  Bennet replied that he had not  But it is  returned she  for
Mrs  Long has just been here  and she told me all about it  Mr  Bennet made no
answer  Do you not want to know who has taken it  cried his wife impatiently
You want to tell me  and I have no objection to hearing it  This was invitation
enough  Why  my dear  you must know  Mrs  Long says that Netherfield is taken by
a young man of large fortune from the north of England  that he came down on
Monday in a chaise and four to see the place  and was so much delighted with it
that he agreed with Mr  Morris immediately  that he is to take possession before
```

Michaelmas and some of his servants are to be in the house by the end of next week What is his name '

Making list from the words in the text

```
[56]: list_of_words = cleaned_multiline_text.split()
      list_of_words
```

```
[56]: ['It',
      'is',
      'a',
      'truth',
      'universally',
      'acknowledged',
      'that',
      'a',
      'single',
      'man',
      'in',
      'possession',
      'of',
      'a',
      'good',
      'fortune',
      'must',
      'be',
      'in',
      'want',
      'of',
      'a',
      'wife',
      'However',
      'little',
      'known',
      'the',
      'feelings',
      'or',
      'views',
      'of',
      'such',
      'a',
      'man',
      'may',
      'be',
      'on',
      'his',
      'first',
      'entering',
```

'a',
'neighbourhood',
'this',
'truth',
'is',
'so',
'well',
'fixed',
'in',
'the',
'minds',
'of',
'the',
'surrounding',
'families',
'that',
'he',
'is',
'considered',
'the',
'rightful',
'property',
'of',
'some',
'one',
'or',
'other',
'of',
'their',
'daughters',
'My',
'dear',
'Mr',
'Bennet',
'said',
'his',
'lady',
'to',
'him',
'one',
'day',
'have',
'you',
'heard',
'that',
'Netherfield',
'Park',

'is',
'let',
'at',
'last',
'Mr',
'Bennet',
'replied',
'that',
'he',
'had',
'not',
'But',
'it',
'is',
'returned',
'she',
'for',
'Mrs',
'Long',
'has',
'just',
'been',
'here',
'and',
'she',
'told',
'me',
'all',
'about',
'it',
'Mr',
'Bennet',
'made',
'no',
'answer',
'Do',
'you',
'not',
'want',
'to',
'know',
'who',
'has',
'taken',
'it',
'cried',
'his',

'wife',
'impatiently',
'You',
'want',
'to',
'tell',
'me',
'and',
'I',
'have',
'no',
'objection',
'to',
'hearing',
'it',
'This',
'was',
'invitation',
'enough',
'Why',
'my',
'dear',
'you',
'must',
'know',
'Mrs',
'Long',
'says',
'that',
'Netherfield',
'is',
'taken',
'by',
'a',
'young',
'man',
'of',
'large',
'fortune',
'from',
'the',
'north',
'of',
'England',
'that',
'he',
'came',

'down',
'on',
'Monday',
'in',
'a',
'chaise',
'and',
'four',
'to',
'see',
'the',
'place',
'and',
'was',
'so',
'much',
'delighted',
'with',
'it',
'that',
'he',
'agreed',
'with',
'Mr',
'Morris',
'immediately',
'that',
'he',
'is',
'to',
'take',
'possession',
'before',
'Michaelmas',
'and',
'some',
'of',
'his',
'servants',
'are',
'to',
'be',
'in',
'the',
'house',
'by',
'the',

```
'end',  
'of',  
'next',  
'week',  
'What',  
'is',  
'his',  
'name']
```

Getting Total length of the list

```
[60]: len(list_of_words)
```

```
[60]: 236
```

Now figure out unique word count from the list

```
[62]: unique_words_as_dict = dict.fromkeys(list_of_words)  
len(list(unique_words_as_dict.keys()))
```

```
[62]: 135
```

```
[65]: for word in list_of_words:  
        if unique_words_as_dict[word] is None:  
            unique_words_as_dict[word] = 1  
        else:  
            unique_words_as_dict[word] += 1  
unique_words_as_dict
```

```
[65]: {'It': 1,  
       'is': 8,  
       'a': 8,  
       'truth': 2,  
       'universally': 1,  
       'acknowledged': 1,  
       'that': 8,  
       'single': 1,  
       'man': 3,  
       'in': 5,  
       'possession': 2,  
       'of': 10,  
       'good': 1,  
       'fortune': 2,  
       'must': 2,  
       'be': 3,  
       'want': 3,  
       'wife': 2,  
       'However': 1,  
       'little': 1,
```

'known': 1,
'the': 8,
'feelings': 1,
'or': 2,
'views': 1,
'such': 1,
'may': 1,
'on': 2,
'his': 5,
'first': 1,
'entering': 1,
'neighbourhood': 1,
'this': 1,
'so': 2,
'well': 1,
'fixed': 1,
'minds': 1,
'surrounding': 1,
'families': 1,
'he': 5,
'considered': 1,
'rightful': 1,
'property': 1,
'some': 2,
'one': 2,
'other': 1,
'their': 1,
'daughters': 1,
'My': 1,
'dear': 2,
'Mr': 4,
'Bennet': 3,
'said': 1,
'lady': 1,
'to': 7,
'him': 1,
'day': 1,
'have': 2,
'you': 3,
'heard': 1,
'Netherfield': 2,
'Park': 1,
'let': 1,
'at': 1,
'last': 1,
'replied': 1,
'had': 1,

'not': 2,
'But': 1,
'it': 5,
'returned': 1,
'she': 2,
'for': 1,
'Mrs': 2,
'Long': 2,
'has': 2,
'just': 1,
'been': 1,
'here': 1,
'and': 5,
'told': 1,
'me': 2,
'all': 1,
'about': 1,
'made': 1,
'no': 2,
'answer': 1,
'Do': 1,
'know': 2,
'who': 1,
'taken': 2,
'cried': 1,
'impatiently': 1,
'You': 1,
'tell': 1,
'I': 1,
'objection': 1,
'hearing': 1,
'This': 1,
'was': 2,
'invitation': 1,
'enough': 1,
'Why': 1,
'my': 1,
'says': 1,
'by': 2,
'young': 1,
'large': 1,
'from': 1,
'north': 1,
'England': 1,
'came': 1,
'down': 1,
'Monday': 1,

```

'chaise': 1,
'four': 1,
'see': 1,
'place': 1,
'much': 1,
'delighted': 1,
'with': 2,
'agreed': 1,
'Morris': 1,
'immediately': 1,
'take': 1,
'before': 1,
'Michaelmas': 1,
'servants': 1,
'are': 1,
'house': 1,
'end': 1,
'next': 1,
'week': 1,
'What': 1,
'name': 1}

```

Now it is interesting and let's get Top 25 words from the unique word list

```

[68]: top_words = sorted(unique_words_as_dict.items(), key=lambda key_val_tuple:
    ↪key_val_tuple[1], reverse=True)
top_words[:25]

```

```

[68]: [('of', 10),
('is', 8),
('a', 8),
('that', 8),
('the', 8),
('to', 7),
('in', 5),
('his', 5),
('he', 5),
('it', 5),
('and', 5),
('Mr', 4),
('man', 3),
('be', 3),
('want', 3),
('Bennet', 3),
('you', 3),
('truth', 2),
('possession', 2),
('fortune', 2),

```

```
('must', 2),
('wife', 2),
('or', 2),
('on', 2),
('so', 2)]
```

3 Activity 2.01

The `itertools` module in Python provides tools for working with iterators. Two functions from this module are `permutations` and `dropwhile`.

```
[71]: from itertools import permutations, dropwhile
      permutations?
      dropwhile?
```

Init signature: `dropwhile(predicate, iterable, /)`

Docstring:

Drop items from the iterable while `predicate(item)` is true.

Afterwards, return every element until the iterable is exhausted.

Type: type

Subclasses:

This function generates all possible arrangements (permutations) of elements from an iterable.

If `r` is specified then it generates permutations of length `r`;

otherwise it generates permutations of the length of the iterable.

```
[76]: permutations(range(3))
```

```
[76]: <itertools.permutations at 0x11d690950>
```

The `isinstance()` function in Python checks if an object belongs to a specific type or a tuple of types.

If the object matches any type within the tuple, it returns `True`; otherwise, it returns `False`.

```
[78]: for number_tuple in permutations(range(3)):
      print(number_tuple)
      assert isinstance(number_tuple, tuple)
```

```
(0, 1, 2)
(0, 2, 1)
(1, 0, 2)
(1, 2, 0)
(2, 0, 1)
(2, 1, 0)
```

```
[81]: for number_tuple in permutations(range(3)):
      print(list(dropwhile(lambda x: x <= 0, number_tuple)))
```

```
[1, 2]
[2, 1]
[1, 0, 2]
[1, 2, 0]
[2, 0, 1]
[2, 1, 0]
```

Convert to number from tuple list above created using permutations

```
[83]: import math
      def convert_to_number(number_stack):
          final_number = 0
          for i in range(0, len(number_stack)):
              final_number += (number_stack.pop() * (math.pow(10, i)))
          return final_number

      for number_tuple in permutations(range(3)):
          number_stack = list(dropwhile(lambda x: x <= 0, number_tuple))
          print(convert_to_number(number_stack))
```

```
12.0
21.0
102.0
120.0
201.0
210.0
```

4 Activity 2.02

```
[87]: from itertools import zip_longest
```

```
[89]: def return_dict_from_csv_line(header, line):
      # Zip them
      zipped_line = zip_longest(header, line, fillvalue=None)
      # Use dict comprehension to generate the final dict
      ret_dict = {kv[0]: kv[1] for kv in zipped_line}
      return ret_dict
```

4.0.1 Reading the data from Sales_record CSV file and return the records until it reaches 10th row.

```
[95]: with open("../DSC540Assignment/sales_record.csv", "r") as fd:
      first_line = fd.readline()
      header = first_line.replace("\n", "").split(",")
      for i, line in enumerate(fd):
```

```

    # Here we loop over the first 10 lines in order to not to make the
↪output too big
    line = line.replace("\n", "").split(",")
    d = return_dict_from_csv_line(header, line)
    print(d)
    if i > 10:
        break

```

```

{'Region': 'Central America and the Caribbean', 'Country': 'Antigua and Barbuda',
 'Item Type': 'Baby Food', 'Sales Channel': 'Online', 'Order Priority': 'M',
 'Order Date': '12/20/2013', 'Order ID': '957081544', 'Ship Date': '1/11/2014',
 'Units Sold': '552', 'Unit Price': '255.28', 'Unit Cost': '159.42', 'Total
Revenue': '140914.56', 'Total Cost': '87999.84', 'Total Profit': '52914.72'}
{'Region': 'Central America and the Caribbean', 'Country': 'Panama', 'Item
Type': 'Snacks', 'Sales Channel': 'Offline', 'Order Priority': 'C', 'Order
Date': '7/5/2010', 'Order ID': '301644504', 'Ship Date': '7/26/2010', 'Units
Sold': '2167', 'Unit Price': '152.58', 'Unit Cost': '97.44', 'Total Revenue':
'330640.86', 'Total Cost': '211152.48', 'Total Profit': '119488.38'}
{'Region': 'Europe', 'Country': 'Czech Republic', 'Item Type': 'Beverages',
 'Sales Channel': 'Offline', 'Order Priority': 'C', 'Order Date': '9/12/2011',
 'Order ID': '478051030', 'Ship Date': '9/29/2011', 'Units Sold': '4778', 'Unit
Price': '47.45', 'Unit Cost': '31.79', 'Total Revenue': '226716.10', 'Total
Cost': '151892.62', 'Total Profit': '74823.48'}
{'Region': 'Asia', 'Country': 'North Korea', 'Item Type': 'Cereal', 'Sales
Channel': 'Offline', 'Order Priority': 'L', 'Order Date': '5/13/2010', 'Order
ID': '892599952', 'Ship Date': '6/15/2010', 'Units Sold': '9016', 'Unit Price':
'205.70', 'Unit Cost': '117.11', 'Total Revenue': '1854591.20', 'Total Cost':
'1055863.76', 'Total Profit': '798727.44'}
{'Region': 'Asia', 'Country': 'Sri Lanka', 'Item Type': 'Snacks', 'Sales
Channel': 'Offline', 'Order Priority': 'C', 'Order Date': '7/20/2015', 'Order
ID': '571902596', 'Ship Date': '7/27/2015', 'Units Sold': '7542', 'Unit Price':
'152.58', 'Unit Cost': '97.44', 'Total Revenue': '1150758.36', 'Total Cost':
'734892.48', 'Total Profit': '415865.88'}
{'Region': 'Middle East and North Africa', 'Country': 'Morocco', 'Item Type':
'Personal Care', 'Sales Channel': 'Offline', 'Order Priority': 'L', 'Order
Date': '11/8/2010', 'Order ID': '412882792', 'Ship Date': '11/22/2010', 'Units
Sold': '48', 'Unit Price': '81.73', 'Unit Cost': '56.67', 'Total Revenue':
'3923.04', 'Total Cost': '2720.16', 'Total Profit': '1202.88'}
{'Region': 'Australia and Oceania', 'Country': 'Federated States of Micronesia',
 'Item Type': 'Clothes', 'Sales Channel': 'Offline', 'Order Priority': 'H',
 'Order Date': '3/28/2011', 'Order ID': '932776868', 'Ship Date': '5/10/2011',
 'Units Sold': '8258', 'Unit Price': '109.28', 'Unit Cost': '35.84', 'Total
Revenue': '902434.24', 'Total Cost': '295966.72', 'Total Profit': '606467.52'}
{'Region': 'Europe', 'Country': 'Bosnia and Herzegovina', 'Item Type':
'Clothes', 'Sales Channel': 'Online', 'Order Priority': 'M', 'Order Date':
'10/14/2013', 'Order ID': '919133651', 'Ship Date': '11/4/2013', 'Units Sold':
'927', 'Unit Price': '109.28', 'Unit Cost': '35.84', 'Total Revenue':
'101302.56', 'Total Cost': '33223.68', 'Total Profit': '68078.88'}

```

```
{'Region': 'Middle East and North Africa', 'Country': 'Afghanistan', 'Item Type': 'Clothes', 'Sales Channel': 'Offline', 'Order Priority': 'M', 'Order Date': '8/27/2016', 'Order ID': '579814469', 'Ship Date': '10/5/2016', 'Units Sold': '8841', 'Unit Price': '109.28', 'Unit Cost': '35.84', 'Total Revenue': '966144.48', 'Total Cost': '316861.44', 'Total Profit': '649283.04'}
{'Region': 'Sub-Saharan Africa', 'Country': 'Ethiopia', 'Item Type': 'Baby Food', 'Sales Channel': 'Online', 'Order Priority': 'M', 'Order Date': '4/13/2015', 'Order ID': '192993152', 'Ship Date': '5/7/2015', 'Units Sold': '9817', 'Unit Price': '255.28', 'Unit Cost': '159.42', 'Total Revenue': '2506083.76', 'Total Cost': '1565026.14', 'Total Profit': '941057.62'}
{'Region': 'Middle East and North Africa', 'Country': 'Turkey', 'Item Type': 'Office Supplies', 'Sales Channel': 'Offline', 'Order Priority': 'C', 'Order Date': '9/25/2013', 'Order ID': '557156026', 'Ship Date': '10/15/2013', 'Units Sold': '3704', 'Unit Price': '651.21', 'Unit Cost': '524.96', 'Total Revenue': '2412081.84', 'Total Cost': '1944451.84', 'Total Profit': '467630.00'}
{'Region': 'Middle East and North Africa', 'Country': 'Oman', 'Item Type': 'Cosmetics', 'Sales Channel': 'Online', 'Order Priority': 'M', 'Order Date': '5/12/2013', 'Order ID': '741101920', 'Ship Date': '5/17/2013', 'Units Sold': '7382', 'Unit Price': '437.20', 'Unit Cost': '263.33', 'Total Revenue': '3227410.40', 'Total Cost': '1943902.06', 'Total Profit': '1283508.34'}
```

[]: